

Sustainability aware asset management

Lecture 6: Climate Change and Climate Risks

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Objectives of the Lecture

In sustainable asset management, there are two main topics:

- how to build a portfolio that satisfies some criteria related to climate change (**decarbonization**)?
 - how to build a portfolio that satisfies some Environmental, Social, or Governance criteria (**ESG investing**)?
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- We start with the presentation of **Climate risks** (Lecture 6)
 - Then, we present the **Portfolio construction under climate risks** (Lectures 7 – 9)
 - Then, we will learn about **ESG investing** (Lectures 10 – 13)

This lecture is related to climate change:

- It summarizes some results related to the “Science of climate change” (IPCC, Intergovernmental Panel on Climate Change)
- It describes these implications of climate change from a finance perspective
 - What is climate risk?
 - How do we measure climate risk in finance?
 - What is the regulation of climate risk?

- **IPCC 6th Assessment Report:** *Climate change 2021 – The Physical Science Basis – Summary for Policymakers*
- **Bolton, P., Despres, M., Pereira Da Silva, L.A., Samama, F., and Svartzman, R.** (2020), *The Green Swan | Central Banking and Financial Stability in the Age of Climate Change*, BIS Publication, <https://www.bis.org/publ/othp31.htm>

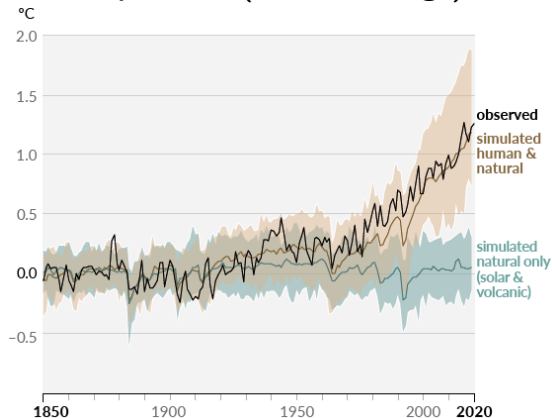
Objectives of the Lecture

- **Climate Change**
- Climate Risk Measurement
- Carbon Data

Global Warming (or Climate Change)

Global warming is the long-term heating of Earth's climate system observed since the pre-industrial period (1850-1900) due to human activities, primarily fossil fuel burning

Change in global surface temperature (annual average) as observed and simulated



Source: IPCC, 6th assessment report, summary for policymakers (2021), page 7.

Some Definitions

Greenhouse gases (GHG) absorb and emit radiation energy, causing the greenhouse effect:

- Water vapor (H_2O)
- Carbon dioxide (CO_2) 74%
- Methane (CH_4) 17%
- Nitrous oxide (N_2O) 6%
- Tropospheric Ozone (O_3)

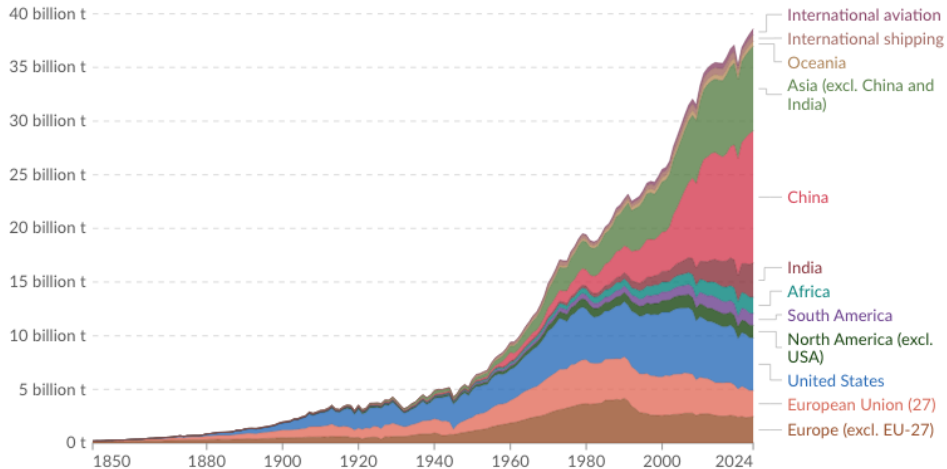
Carbon dioxide equivalent (or CO_2e) is a term describing different GHG in a common unit

- A quantity of GHG can be expressed as CO_2e by multiplying the amount of the GHG by its global warming potential over a period of 100 years
- 1 kg of methane corresponds to 23 kg of CO_2 (lifetime: 12 years)
- 1 kg of nitrous oxide corresponds to 296 kg of CO_2 (lifetime: 121 years)

Source: IPCC.

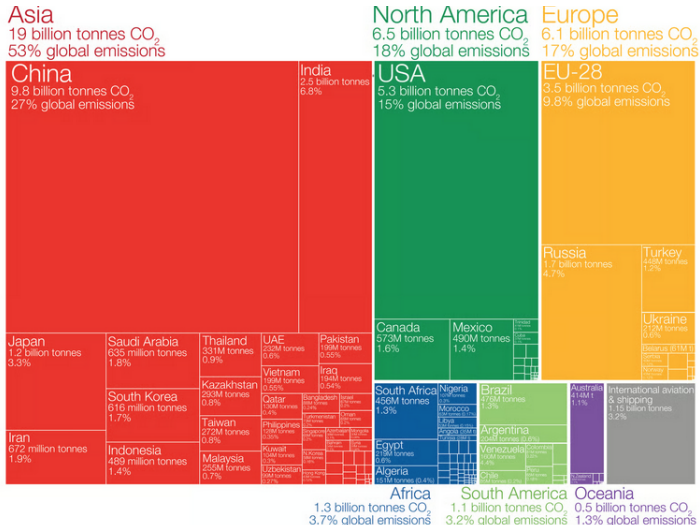
Annual CO₂ Emissions by Region

CO₂ emissions from the burning of fossil fuels for energy and cement production



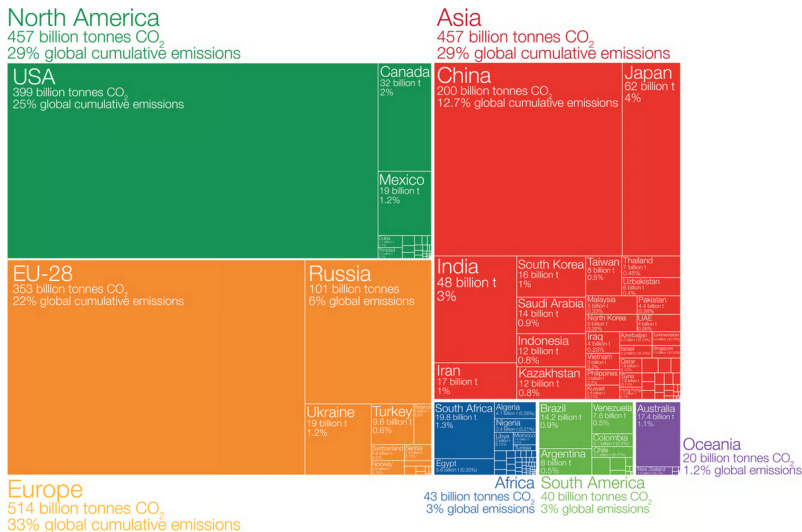
Source: Global Carbon Project (<https://ourworldindata.org/co2-and-other-greenhouse-gas-emissions>)

CO₂ Emissions in 2017 (36.2 billion tonnes in total)



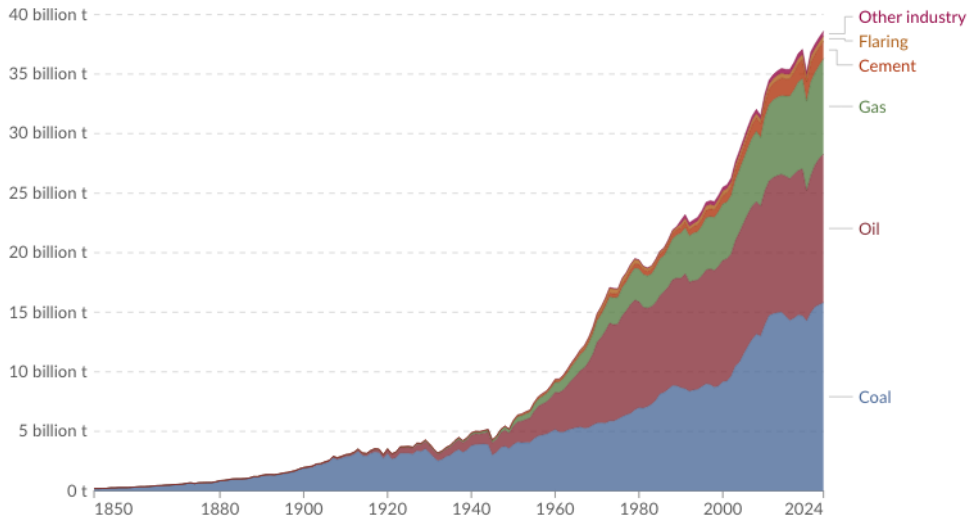
Source: Global Carbon Project (<https://ourworldindata.org/co2-and-other-greenhouse-gas-emissions>)

Cumulative CO₂ Emissions (1.5 trillion tonnes since 1751)



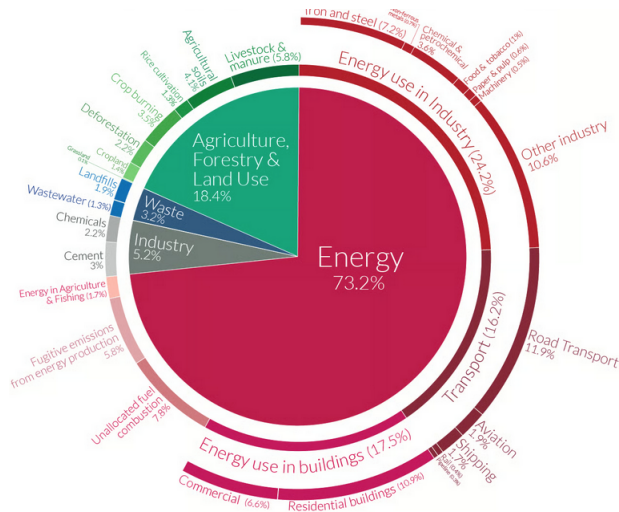
Source: Global Carbon Project (<https://ourworldindata.org/co2-and-other-greenhouse-gas-emissions>)

Source of CO2 Emissions by Fuel Type (World)



Source: Global Carbon Project (<https://ourworldindata.org/co2-and-other-greenhouse-gas-emissions>)

GHG Emissions by Sector (World, 2016) (49.4 billion tonnes CO₂e)

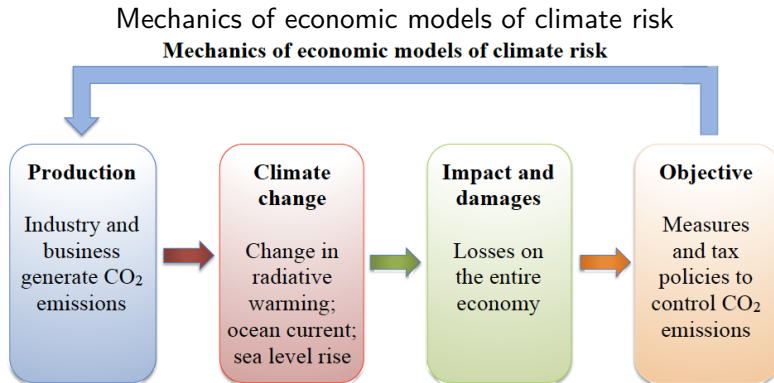


Source: Climate Watch, the World Resources Institute (2020) (<https://ourworldindata.org/emissions-by-sector>)

Modeling the Interaction between Climate and Economy

Integrated Assessment Models (IAM) have 3 components

- Economic growth
- Climate dynamics
- An objective function



Modeling the Interaction between Climate and Economy

The most famous IAM is the Dynamic Integrated model of Climate and the Economy (or DICE) developed by Nordhaus (1993, 2018). Sketch of the model:

Step 1: Production results in CO₂ emissions $E_t = f(Y_t)$ with Y_t : gross output
CO₂ accumulation results in CO₂ concentration $C_t = \phi_C C_{t-1} + \beta_C E_t$

Step 2: CO₂ concentration results in increase in radiative forcing $F_t = g(C_t) + F_{\text{Exo},t}$
Radiative forcing affects temperature $T_t = \phi_T T_{t-1} + \beta_T F_t$

Step 3: Increase in temperature generates climate damages $D_t = a_1 T_t + a_2 T_t^2$
Impact of damages can be attenuated by an abatement technology at a cost $\Lambda_t = b_1 \mu_t^{b_2}$ with μ_t % reduction in CO₂ emissions: $E_t = \sigma_t (1 - \mu_t) Y_t$
(σ_t is the exogenous reduction in CO₂ intensity of production)

Step 4: Damage and abatement cost results in a loss of production so that net output is $Q_t = (1 - D_t)(1 - \Lambda_t) Y_t$
Production is reduced because of the damages and because a fraction of the production is devoted to reducing CO₂ emissions

Modeling the Interaction between Climate and Economy

Difficulty: Calibration of the model parameters because the model is highly nonlinear and non-stationary

In fact, the climate block is more complicated (temperature for atmosphere, upper ocean, and lower ocean). Emissions flow into the atmosphere but ocean can capture part of the emissions.

Applications:

- Compare baseline scenario of the different models
- Compute the 2°C scenario, the optimal welfare scenario, etc.
- Compute the social cost of carbon (SCC): optimal tax rate for carbon emissions.

SCC: Economic cost caused by an additional tonne of carbon dioxide emissions (no consensus)

- \$266/tCO₂ for Stern (2007)
- \$57/tCO₂ for Golosov et al. (2014)
- \$31.2/tCO₂ for Nordhaus (2018) in optimal welfare
- \$229/tCO₂ for Nordhaus (2018) in 2.5°C scenario

IPCC Projections

The **Intergovernmental Panel on Climate Change (IPCC)** is the United Nations body for assessing the science related to climate change (GIEC in French)

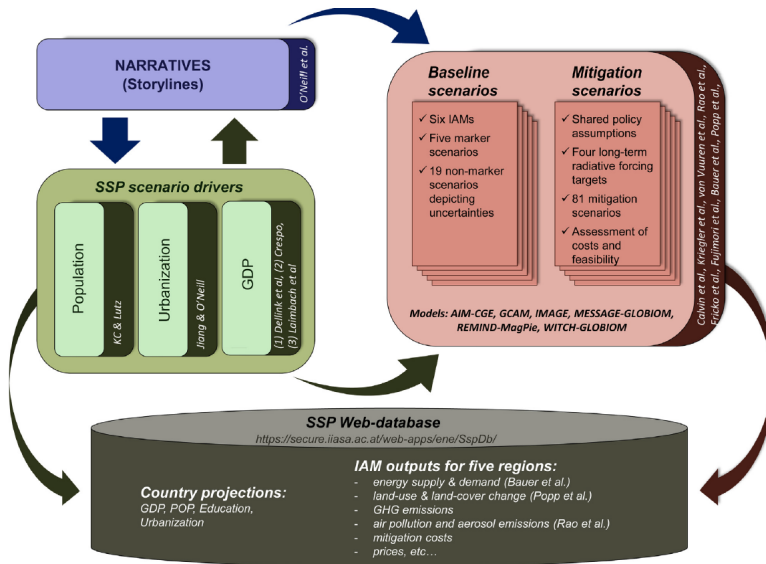
The IPCC was created to provide policymakers with regular scientific assessments on climate change, its implications and potential future risks, as well as to put forward adaptation and mitigation options

Working Groups:

- Working Group I (WGI): physical science underpinning past, present, and future climate change
- Working Group II (WGII): impacts, adaptation and vulnerabilities related to climate change
- Working Group III (WGIII): climate change mitigation, assessing methods for reducing greenhouse gas emissions, and removing greenhouse gases from the atmosphere

2021–2022: 6th Assessment Report (<https://www.ipcc.ch/assessment-report/ar6/>)

IPCC Assessment Report 6 (2021)

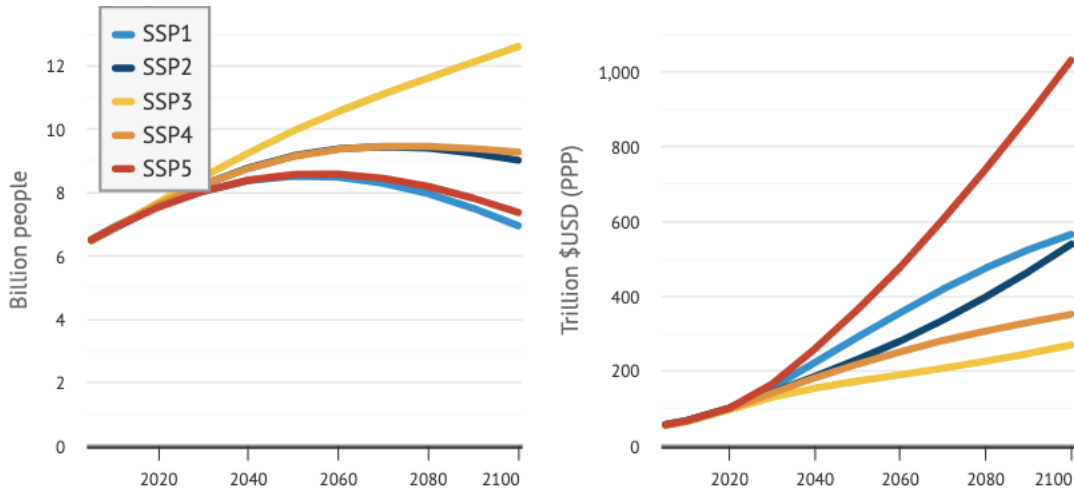


IPCC AR6: Possible Climate Futures

Scenarios are called **Shared socioeconomic pathways (SSP)**

- **SSP1 Sustainability** – Taking the Green Road (Low challenges to mitigation and adaptation) The world shifts toward a more sustainable path.
- **SSP2 Middle of the Road** (Medium challenges to mitigation and adaptation) The world follows a path in which social, economic, and technological trends do not shift markedly from historical patterns.
- **SSP3 Regional Rivalry** – A Rocky Road (High challenges to mitigation and adaptation) A resurgent nationalism, concerns about competitiveness and security, and regional conflicts push countries to increasingly focus on domestic and regional issues.
- **SSP4 Inequality** – A Road Divided (Low challenges to mitigation, high challenges to adaptation) Highly unequal investments in human capital and increasing disparities in economic opportunity and political power, lead to increasing inequalities both across and within countries.
- **SSP5 Fossil-fueled Development** – Taking the Highway (High challenges to mitigation, low challenges to adaptation) This world places increasing faith in competitive markets, innovation and participatory societies to produce rapid technological progress and development of human capital as the path to sustainable development.

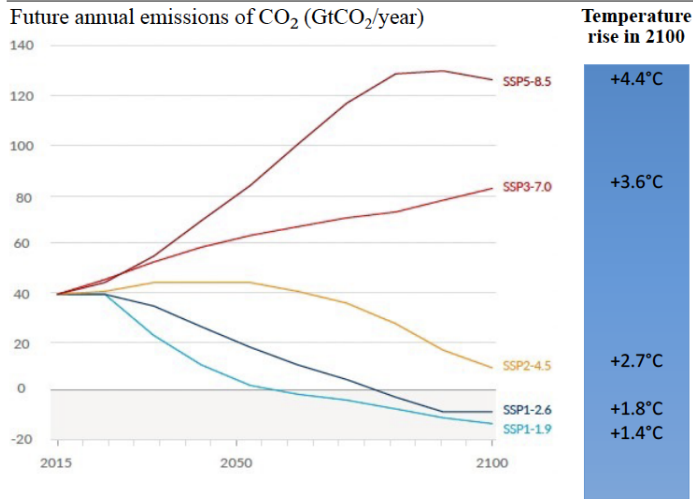
IPCC AR6: Projections of population and economic growth



Source:

<https://www.carbonbrief.org/explainer-how-shared-socioeconomic-pathways-explore-future-climate-change>

IPCC AR6: Projections of Carbon Emissions



SSPx-y:

- x: number of the scenario
- y: total radiative forcing (in W/m²) by 2100

Radiative forcing

measures the change in the energy balance of the Earth's climate system caused by certain factors, primarily greenhouse gases and aerosols.

Source: IPCC, 6th assessment report, summary for policymakers (2021), page 16.

IPCC AR6: Projections of Temperature

Changes in global surface temperature. Temperature differences relative to the average global surface temperature of the period 1850-1900 are reported in °C.

Scenario	Near term, 2021–2040		Mid-term, 2041–2060		Long term, 2081–2100	
	Best estimate (°C)	<i>Very likely</i> range (°C)	Best estimate (°C)	<i>Very likely</i> range (°C)	Best estimate (°C)	<i>Very likely</i> range (°C)
SSP1-1.9	1.5	1.2 to 1.7	1.6	1.2 to 2.0	1.4	1.0 to 1.8
SSP1-2.6	1.5	1.2 to 1.8	1.7	1.3 to 2.2	1.8	1.3 to 2.4
SSP2-4.5	1.5	1.2 to 1.8	2.0	1.6 to 2.5	2.7	2.1 to 3.5
SSP3-7.0	1.5	1.2 to 1.8	2.1	1.7 to 2.6	3.6	2.8 to 4.6
SSP5-8.5	1.6	1.3 to 1.9	2.4	1.9 to 3.0	4.4	3.3 to 5.7

Source: IPCC, 6th assessment report, summary for policymakers (2021), page 14

IPCC AR6: Remaining Carbon Budget

Estimates of historical CO₂ emissions and remaining carbon budgets. Estimated remaining carbon budgets are calculated from the beginning of 2020 and extend until global net zero CO₂ emissions are reached.

Global Warming Between 1850–1900 and 2010–2019 (°C)		Historical Cumulative CO ₂ Emissions from 1850 to 2019 (GtCO ₂)					
1.07 (0.8–1.3; likely range)		2390 (± 240; likely range)					
Approximate global warming relative to 1850–1900 until temperature limit (°C) ^a	Additional global warming relative to 2010–2019 until temperature limit (°C)	Estimated remaining carbon budgets from the beginning of 2020 (GtCO ₂)					Variations in reductions in non-CO ₂ emissions ^c
		<i>Likelihood of limiting global warming to temperature limit^b</i>					
		17%	33%	50%	67%	83%	
1.5	0.43	900	650	500	400	300	Higher or lower reductions in accompanying non-CO ₂ emissions can increase or decrease the values on the left by 220 GtCO ₂ or more
1.7	0.63	1450	1050	850	700	550	
2.0	0.93	2300	1700	1350	1150	900	

Source: IPCC, 6th assessment report, summary for policymakers (2021), page 29

IPCC AR6: Carbon Dioxide Removal (CDR)

Anthropogenic CO₂ removal (CDR) has the potential to remove CO₂ from the atmosphere and durably store it in reservoirs. CDR aims to compensate for residual emissions to reach net zero CO₂ (or GHG) emissions or, if implemented at a scale where anthropogenic removals exceed anthropogenic emissions, to lower surface temperature.

Nature-based solutions

- Afforestation
- Reforestation
- Restoration of peat bogs
- Restoration of coastal and marine habitats

Enhanced natural processes

- Land management and no-till agriculture, which avoids carbon release through soil disturbance
- Better wildfire management
- Ocean fertilization to increase its capacity to absorb CO₂

Technology solutions

- Bioenergy with carbon capture and storage (BECCS)
- Direct air capture (DAC)

Objectives of the Lecture

- Climate Change
- **Climate Risk Measurement**
- Carbon Data

Climate risks and financial losses

Climate Risks include physical risks and transition risks:

Physical risks represent the economic costs and financial losses due to increasing frequency and severity of climate-related weather events (e.g., storms, floods or heat waves) and the effects of long-term changes in climate patterns (e.g., ocean acidification, rising sea levels or changes in precipitation).

Transition risks are associated with the uncertain financial impacts that could result from a rapid low-carbon transition, including policy changes, reputational impacts, technological breakthroughs, or limitations, and shifts in market preferences and social norms.

A rapid and ambitious transition to lower emissions pathways means that a large fraction of proven reserves of fossil fuel cannot be extracted, becoming “stranded assets”, with potentially systemic consequences for the financial system.

Remark: A third type of risk, **liability risk**, refers to the impacts that could arise tomorrow if parties who have suffered loss or damage from the effects of climate change seek compensation from those they hold responsible.

Climate risks and financial losses

Physical Risk

- More an operational risk than a business risk
- Measuring physical risk is a difficult task
- Strong impact on real estate and insurance sectors
- Probably low impact on stock prices

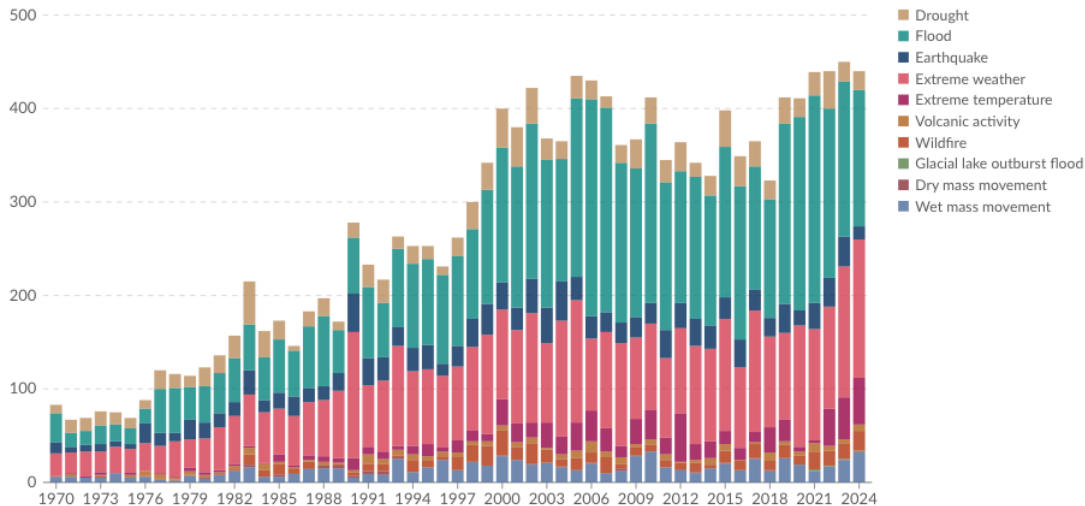
Transition Risk

- A business risk
- Measuring transition risk is a difficult task
- Impact on many sectors (energy, utilities, materials)
- Probably high impact on stock prices



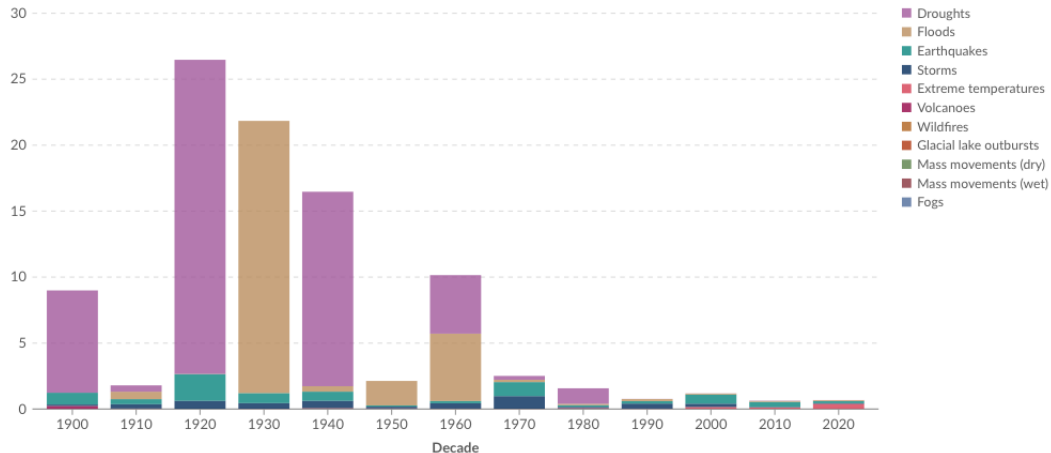
Source: BIS (2021), page 19

Physical Risks: Number of Reported Natural Disasters



Source: EMDAT (2022). <https://ourworldindata.org/grapher/natural-disasters-by-type>

Annual Deaths from Natural Disasters (per 100,000 people)



Source: EMDAT (2022). https://ourworldindata.org/grapher/decadal-average-death-rates-from-natural-disasters?country=~OWID_WRL

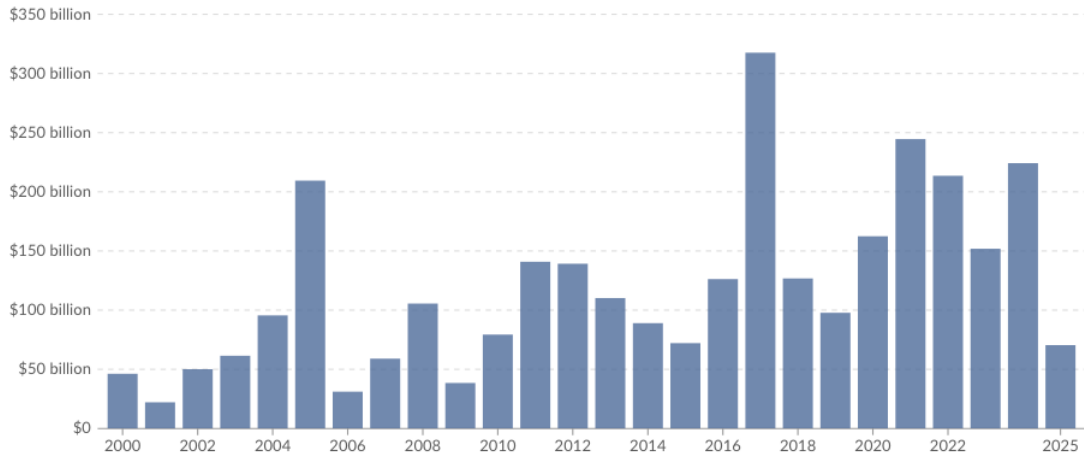
https://ourworldindata.org/grapher/decadal-average-death-rates-from-natural-disasters?country=~OWID_WRL

Physical Risks: Climate Disasters

	1960–2019	1960–1979	1980–1999	2000–2019
Average number of climate-related disasters per year				
Floods	84.0	20.9	69.5	161.5
Storms	66.6	25.2	72.9	101.8
Landslides	11.8	4.0	12.7	18.7
Wildfires	7.3	1.8	8.2	11.9
Heat waves	3.4	0.8	2.4	7.0
Cold waves	6.3	0.5	4.2	14.4
Droughts	11.9	5.6	13.2	16.8
Average number of deaths per disaster				
Floods	107	493	155	37
Storms	481	2585	274	108
Landslides	102	367	89	55
Wildfires	8	6	10	7
Heat waves	858	197	277	1124
Cold waves	72	306	97	58
Droughts	8509	40088	5828	82

Source: EMDAT (2020)

Physical Risks: Financial Losses from Climate Disasters



Source: EMDAT (2022). <https://ourworldindata.org/grapher/damage-costs-from-natural-disasters?time=2000..latest&country=~All+disasters+excluding+earthquakes>

Transition Risks

Transition risks are relevant to financial institutions holding at least two types of financial sets (representing about 30% of global equity and fixed-income investments):

- **Tier 1** – Securities of firms that may be impacted directly by regulatory limits in terms of their ability to produce or use fossil fuels (e.g., coal, oil and gas, and extraction companies) (**Stranded Assets**)
- **Tier 2** – Securities of firms that are energy-intensive (e.g., forestry, paper, metals, and mining, etc.).

A good measure of transition risk is **carbon risk**.

Transition Risks: Stranded Assets

Limiting global warming to less than 1.5°C or 2°C requires keeping a large proportion of existing fossil fuel reserves in the ground. They are called **Stranded Assets**.

- These assets have suffered or may suffer from unanticipated or premature write-downs, devaluations or conversion to liabilities
- For example, a 2°C alignment implies to keep a large proportion of existing fossil fuel reserves in the ground (around 30% of oil reserves, 50% of gas reserves and 80% of coal)
- Risk factors: Regulations, carbon prices, change in demand, social pressure, etc.
- The discounted loss in global wealth resulting from stranded fossil fuel assets may range from \$1 trillion to \$4 trillion (Mercure et al., 2018), far below the International Renewable Energy Agency's (IRENA) (2017) estimate of \$18 trillion.

Objectives of the Lecture

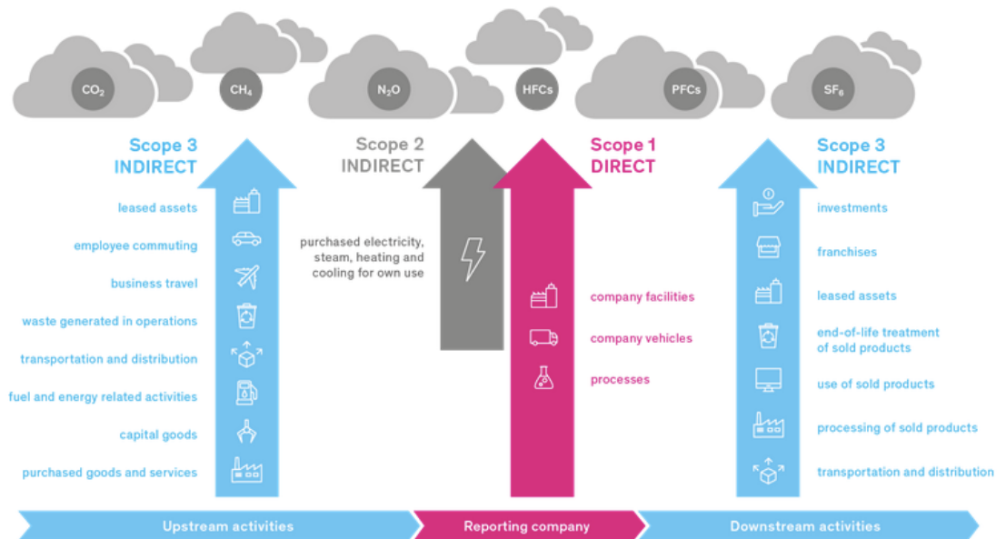
- Climate Change
- Climate Risk Measurement
- **Carbon Data**

The GHG Protocol corporate standard classifies a company's GHG emissions in three scopes:

- **Scope 1:** GHG emissions generated from burning fossil fuels and production processes which are owned or controlled by the company (direct emissions).
- **Scope 2:** GHG emissions from consumption of purchased electricity, heat, or steam by the company (first-tier indirect emissions).
- **Scope 3:** Other indirect GHG emissions that occur in the value chain of the reporting company, including both upstream and downstream emissions, such as from the extraction and production of purchased materials and fuels, transport-related activities in vehicles not owned or controlled by the reporting entity, electricity-related activities not covered in Scope 2, outsourced activities, waste disposal, etc.

Remark: Scopes 1 and 2 are mandatory to report, whereas scope 3 is voluntary (and harder to measure and monitor).

Carbon Indicators



Source: <https://www.myclimate.org>

Carbon Indicators

Carbon emissions: measured in tonnes of CO2 equivalent (tCO2e)

Carbon intensity: amount of GHG emissions per unit of revenues. It is measured in tonnes of CO2 equivalent per dollar of revenue (tCO2e/\$ mln).

How to find data of carbon emission and intensity?

Carbon Disclosure Project (CDP) is a not-for-profit charity that runs the global disclosure system for investors, companies, cities, states, and regions to manage their environmental impacts (<https://www.cdp.net>)

Trucost was established to provide the data, tools and insights needed by companies, investors and policy makers to deliver the transition to a low carbon, resource efficient economy (<https://www.trucost.com>)

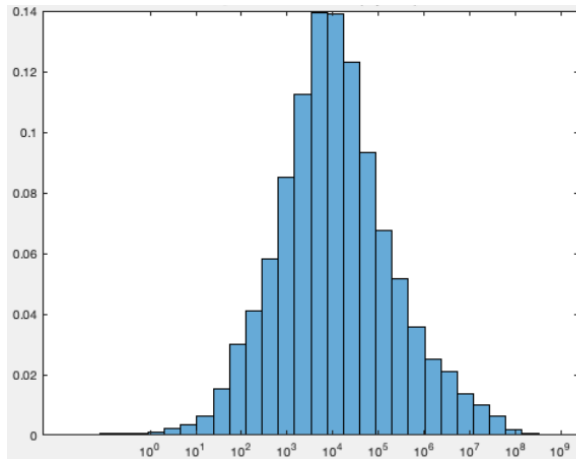
Carbon Data – Examples of Scope 1–3 Carbon Emissions and Intensity (2019)

	Walmart (Retail)	Southern Company (Elec. prod.)	Exxon Mobil (Oil&gas prod.)
Scope 1 emissions (mln tonnes)	6.1	88.2	111.0
Scope 2 emissions (mln tonnes)	13.1	3.4	9.0
Scope 3 emissions (mln tonnes)	40.7	5.0	107.3
Revenue (bln \$)	514.4	20.7	255.6
Scope 1 intensity (tonnes/\$ mln rev.)	11.9	4269.3	434.3
Scope 2 intensity (tonnes/\$ mln rev.)	25.4	165.7	35.2
Scope 3 intensity (tonnes/\$ mln rev.)	79.0	240.0	419.8
Scope 1–3 emissions (mln tonnes)	59.8	96.6	227.3
Scope 1–3 intensity (tonnes/\$ mln rev.)	116.3	4673.8	889.3

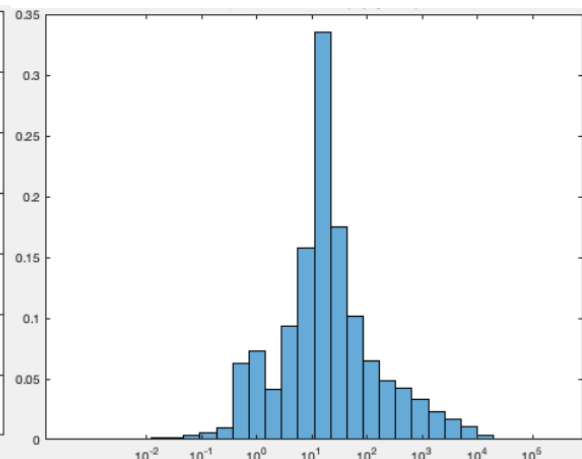
Source: Trucost (2021).

Distribution of Carbon Emissions and Intensity (2019)

Scope 1 Carbon Emissions (tCO₂e)



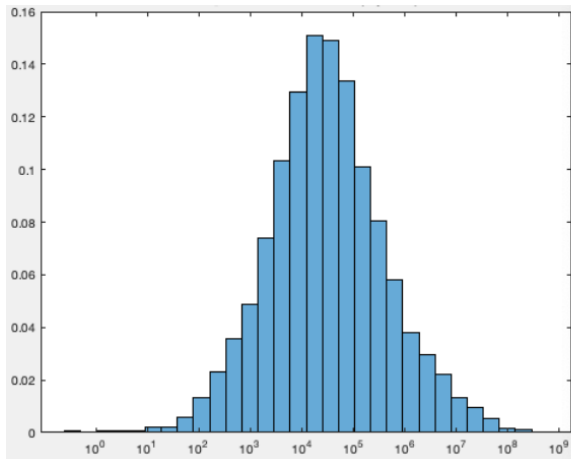
Scope 1 Carbon Intensity (tCO₂e/\$ mln)



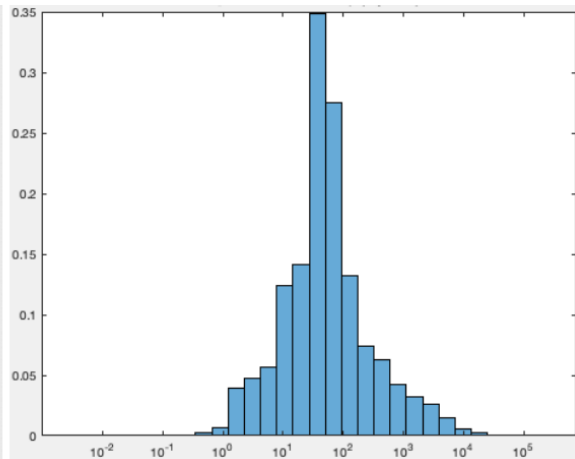
Source: Trucost (2021)

Distribution of Carbon Emissions and Intensity (2019)

Scope 1-2 Carbon Emissions (tCO₂e)



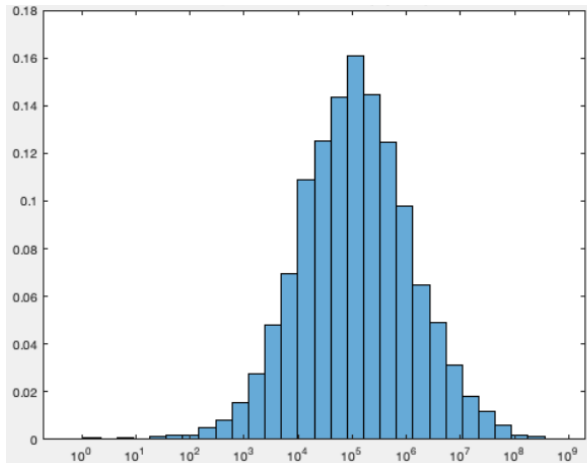
Scope 1-2 Carbon Intensity (tCO₂e/\$ mln)



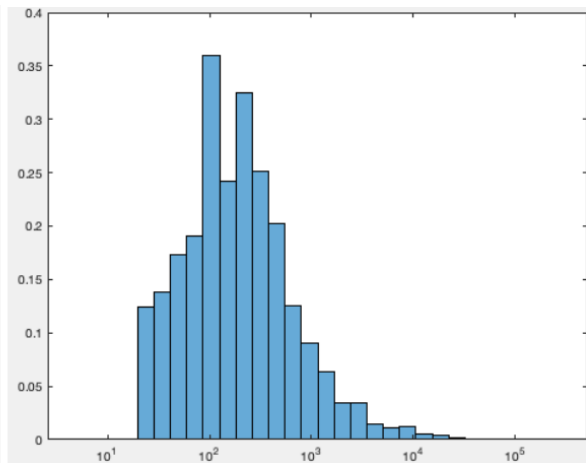
Source: Trucost (2021)

Distribution of Carbon Emissions and Intensity (2019)

Scope 1-3 Carbon Emissions (tCO₂e)



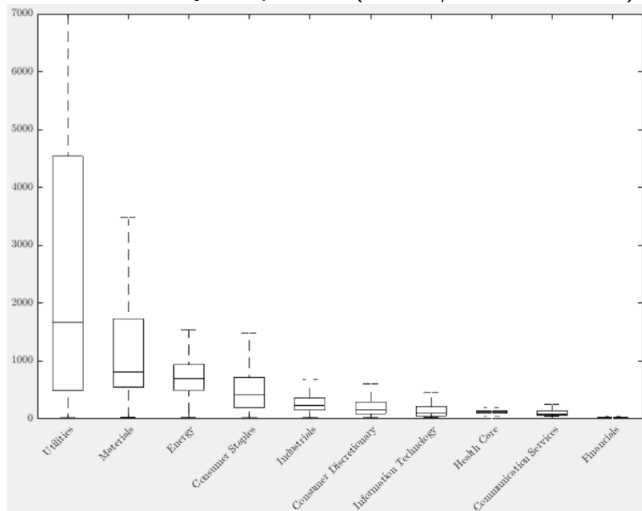
Scope 1-3 Carbon Intensity (tCO₂e/\$ mln)



Source: Trucost (2021)

Distribution of Carbon Intensity by Sector (2019, World)

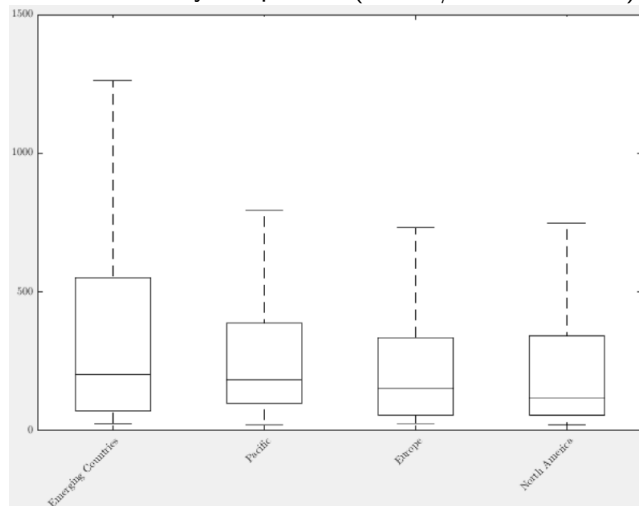
Carbon intensity Scope 1–3 (tCO₂e/\$ mln revenue)



Remark: Box represents the IQR range. Max and min represent $Q3 + 1.5 \text{ IQR}$ and $Q1 - 1.5 \text{ IQR}$, respectively.

Distribution of Carbon Intensity by Region (2019)

Carbon intensity Scope 1–3 (tCO₂e/\$ mln revenue)



Remark: Box represents the IQR range. Max and min represent $Q3 + 1.5 \text{ IQR}$ and $Q1 - 1.5 \text{ IQR}$, respectively.

Which Scope to Use?

Scopes 1 and 2 are mandatory to report and should be included in any portfolio construction

Scope 3: It matters for some industries

- Banks emit little carbon directly, but they finance polluting firms (Scope 3)
- Car producers have a large influence on the emissions of their cars (Tesla vs. Ford)

Problem: most companies do not disclose Scope 3 emissions. They are estimated by data providers (CDP, Trucost) but estimates include some uncertainty.

Sectors for which Scope 3 is likely to be material

Upstream Scope 3:

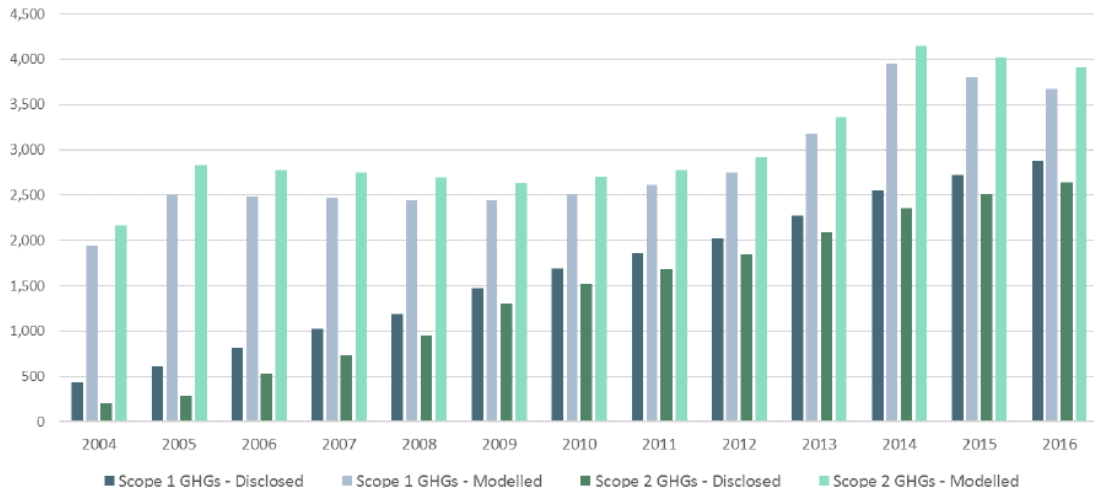
- Energy, utilities, materials, chemicals, industrials, consumer goods

Downstream Scope 3:

- Sold products: oil and gas, automobiles, technology, apparel, chemicals
- Financed emissions: financials

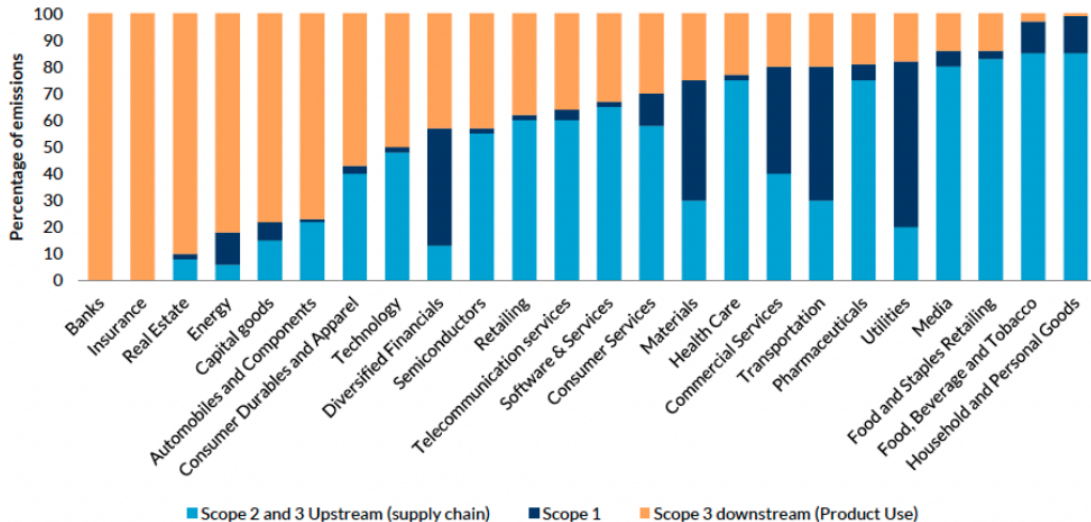
Which Scope to Use?

Historical GHG Data Disclosure Trends (Core Universe) - *Disclosed vs. Modelled*



Source: Trucost (2019)

Which Scope to Use?



Source: Based on Inrate data

Double Counting

Double counting: the same tonne of carbon is counted multiple times within a portfolio

Category	Scope	Example
Electricity producer/user	Scope 2	Company A produces the electricity (Scope 1) purchased by company B (Scope 2)
Supplier/user	Scope 3	Company A uses the energy-consuming goods (Scope 1) produced by company B (Scope 3)
Product/energy	Scope 3	The petrol produced by company A (Scope 3) is burnt by the vehicles operated by (Scope 1) or produced by (Scope 3) company B
Producer/retailer	Scope 3	Company A is a retailer (product in use Scope 3) of goods purchased by company B (Scope 3)

- Large and diversified portfolios are more prone to double counting.
- In principle, the same emissions could be counted 3 times
- Eliminating all double counting is virtually impossible (requires input-output models)
- It may reach up to 30–40% of total emissions.

End of Lecture 6